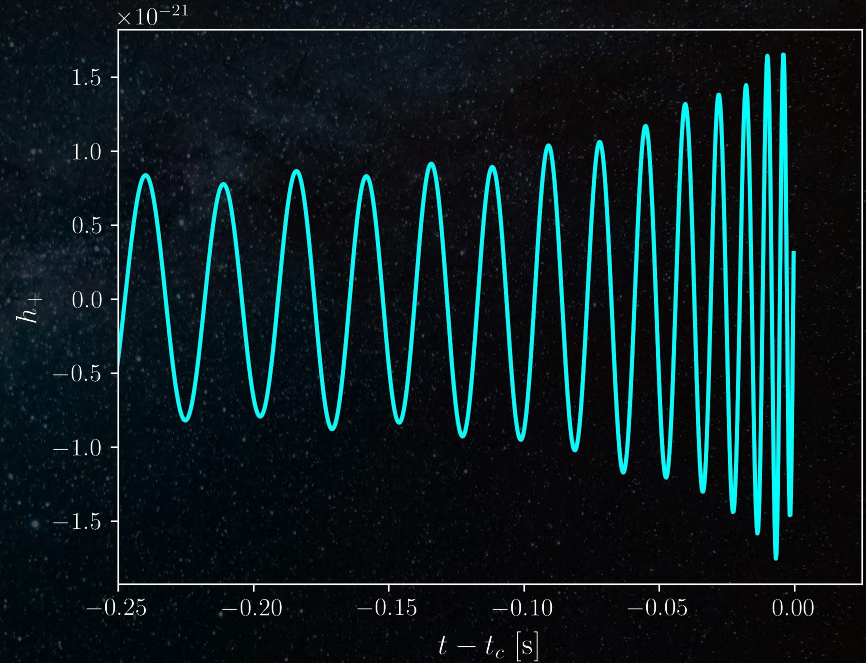


Including Gravitational Waveform Uncertainty in Bayesian Inference of Source Properties

Ryan Johnson, Dr. Jocelyn Read

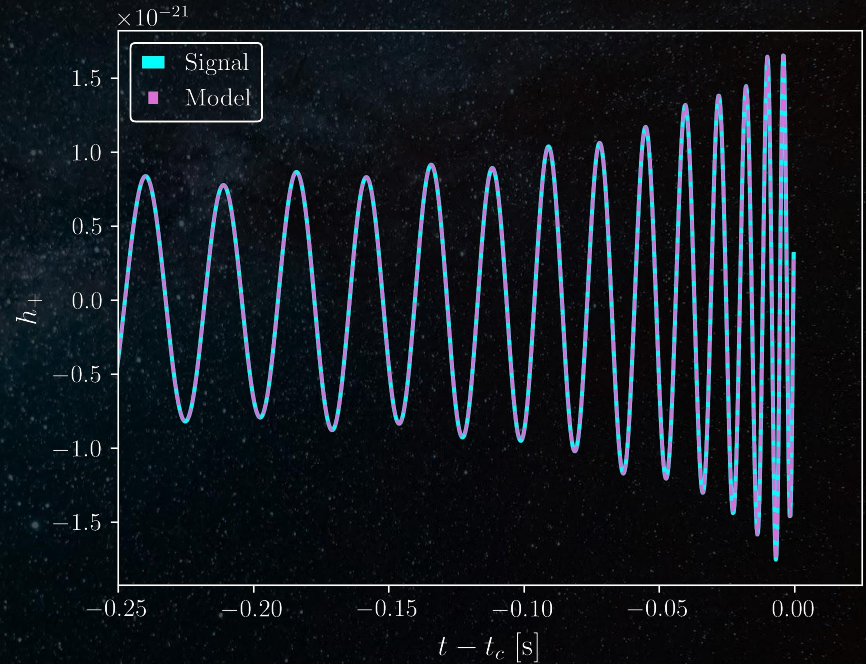
Compact Binary Inspiral Gravitational Waves

- Produced by binary black holes and neutron stars
- Measured by interferometers
 - LIGO, Virgo, Kagra
- Signals received as gravitational wave strain, h



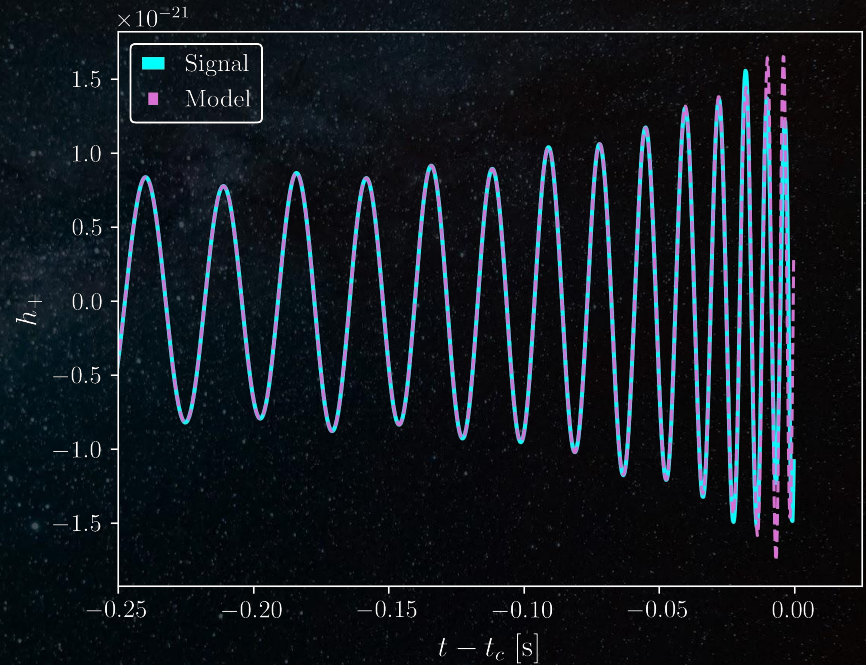
Parameter Estimation

- Signals are analyzed by fitting waveform models
 - Bayesian Inference
 - Generate a model, compare to recorded data
- Outputs posteriors
 - Probability distributions of the source parameters



Waveform Systematics

- Usually: assume the model is capable of accurately representing the signal for given parameters
- If the model cannot fully match the signal, we get bias in the posterior distributions
 - **Waveform Uncertainty**



Waveform Corrections

Frequency domain strain $\tilde{h}$ comes from model μ with parameters ϑ

$$\tilde{h}(f) \approx \mu(f; \vartheta)$$

If the model is flawed, the strain will have amplitude and phase differences:

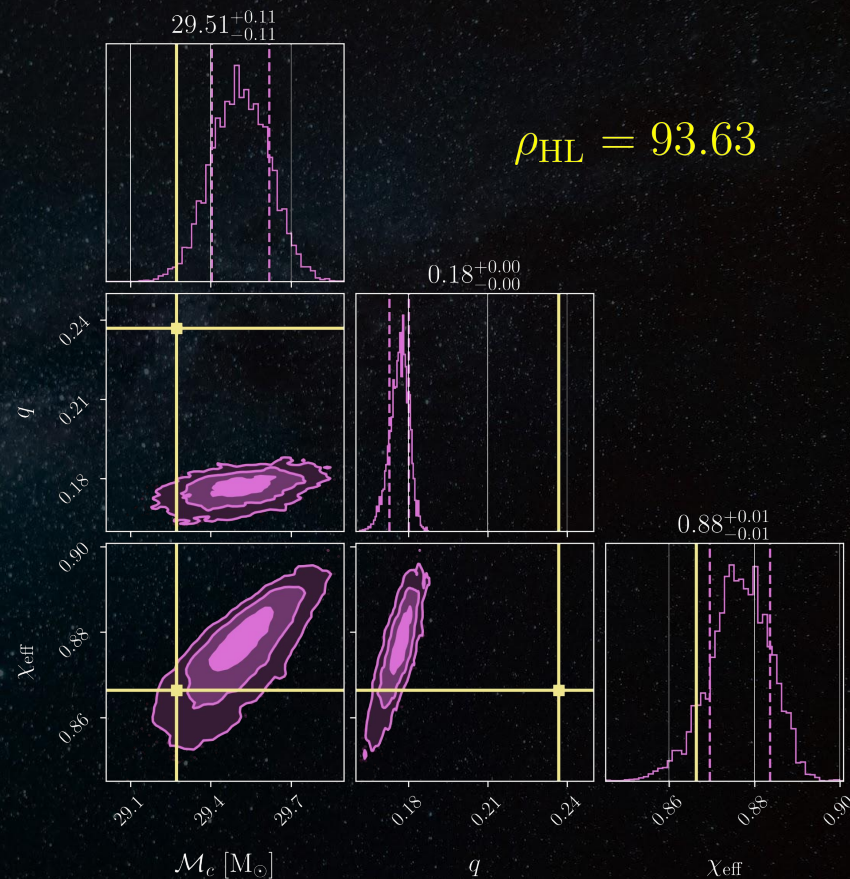
$$\tilde{h}(f) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}(f)) \exp(i\Delta\phi(f))$$

Captures corrections to amplitude and phase:

$$\Delta\mathcal{A}(f) = \frac{\mathcal{A}_{\text{true}}(f)}{\mathcal{A}_{\text{model}}(f)} - 1 \quad \Delta\phi(f) = \phi_{\text{true}}(f) - \phi_{\text{model}}(f)$$

Impact of Systematics

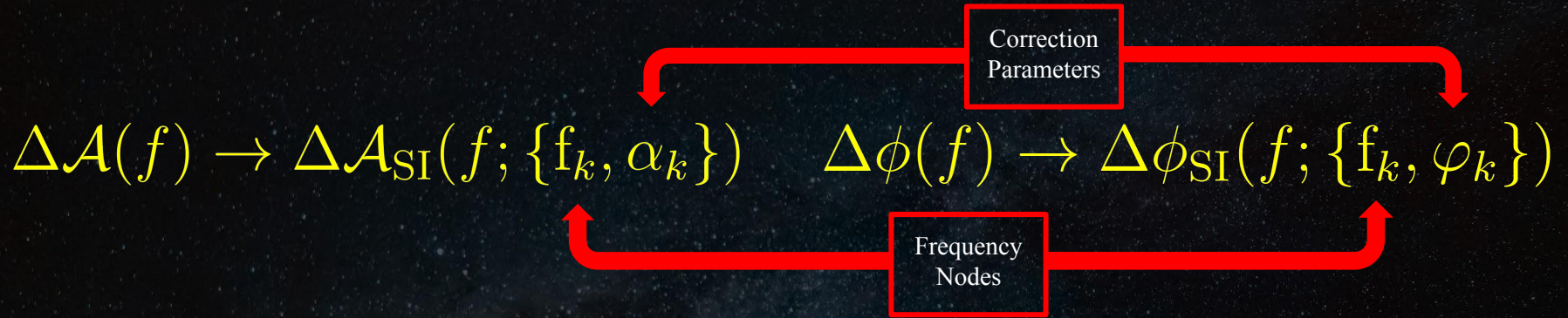
- Construct “corrected” signal from model plus a phase deviation
- $\text{match}(\mu(f; \vartheta), \mu_c(f; \vartheta, \Phi)) = 98.1\%$
- Perform parameter estimation with the original model
- Model has to compensate for the difference using the source parameters
- We see significant bias in recovered values



How do we fix it?

- We additionally infer corrections in the amplitude and phase of a waveform model
 - These corrections must be small to not fully override general relativity model
 - We don't assume a specific form of correction
- With the corrections allowed, any systematics in the model should be captured by it during parameter estimation, reducing the bias in the posteriors.

Correction Model



$$\mu_c(f; \vartheta, \alpha, \varphi) = \mu(f; \vartheta)(1 + \Delta\mathcal{A}_{\text{SI}}(f; \{f_k, \alpha_k\})) \exp(i\Delta\phi_{\text{SI}}(f; \{f_k, \varphi_k\}))$$

Range of Corrections

- Assume that waveform systematics are most likely to occur
 - past the early inspiral regime of the waveform
 - End of PN-only region in **IMR** models
 - Khan et. al. Phys. Rev. D 93, 044007
 - before the light ring frequency
- Place nodes within a range of dimensionless frequency

$$\xi \equiv \frac{GMf}{c^3}$$

$$\xi_{\min} = \xi_{\text{IM}} = 0.018$$

$$\xi_{\max} = 2\xi_{\text{light}} = 1/\pi$$

Variable Frequency Nodes

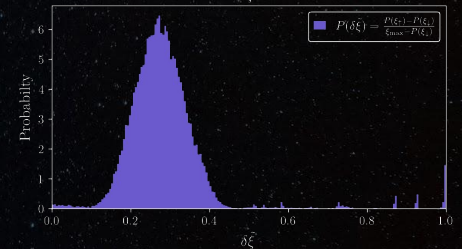
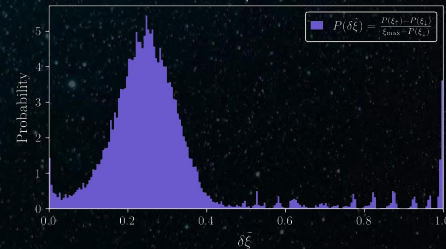
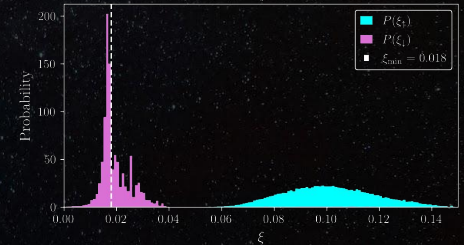
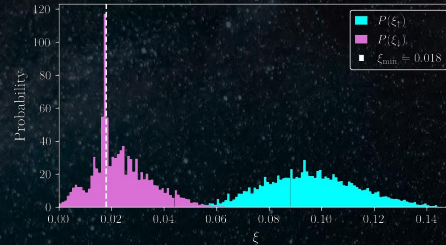
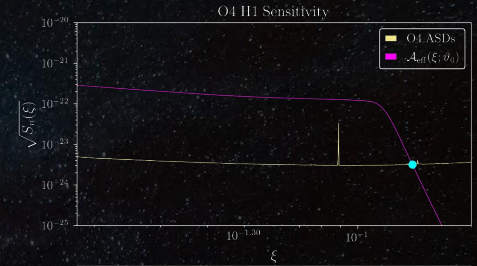
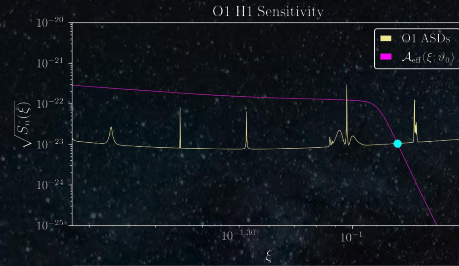
- We allow the code to choose:
 - Where the correction ‘starts’: $\xi_0 \in [\xi_{\min}, \xi_{\max})$
 - How the nodes are spaced: $\delta\tilde{\xi} \in (0, 1]$
- k th node of $n+1$ nodes is given by:

$$f_k = \frac{c^3 \xi_0}{GM} \left(1 + \left[\frac{\xi_{\max} - \xi_0}{\xi_0} \right] \delta\tilde{\xi} \right)^{k/n}$$

Prior on Nodes

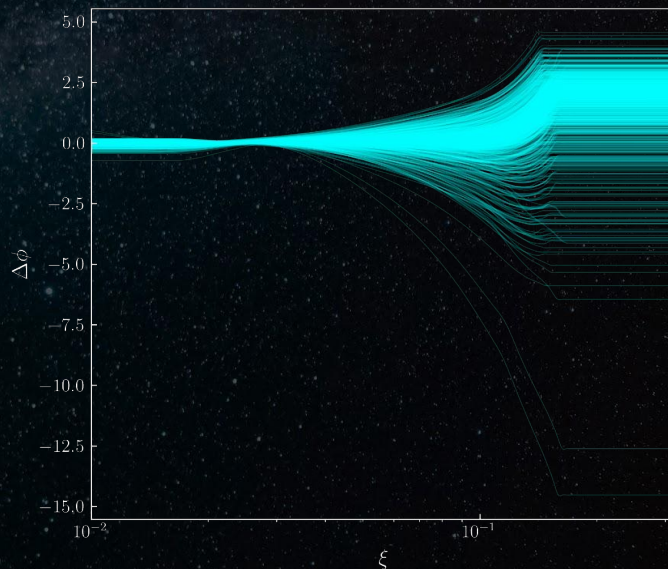
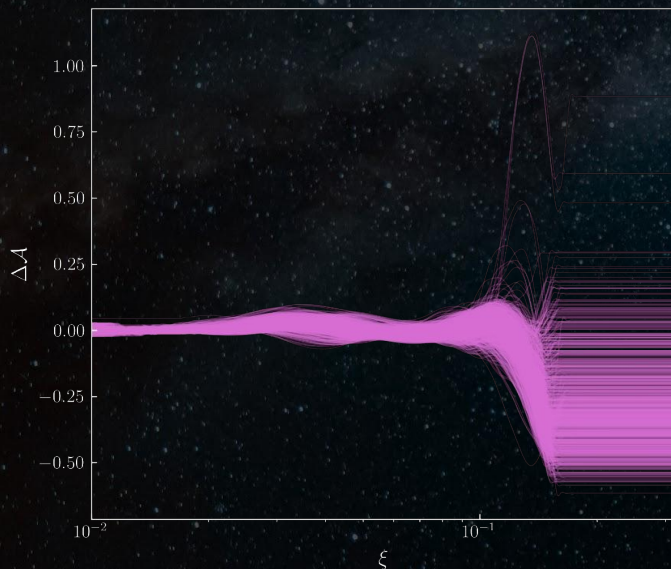
$$\sqrt{\frac{4c^3\xi}{GM}}|\mu(\xi;\vartheta)| - \sqrt{S_n(\xi)} = 0$$

- Solutions to this equation tell us where the waveform crosses the noise
- Construct priors for the starting frequency node and spacing



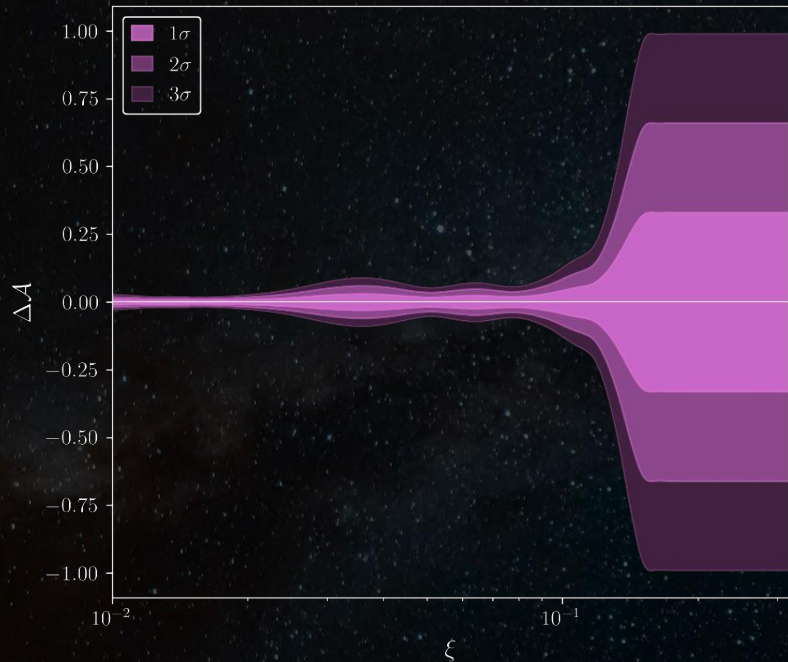
Correction Parameter Priors

- Motivated by the amplitude and phase differences between two waveform models
 - **IMRPhenomD** and **SEOBNRv4**
 - Draw thousands of samples of source parameters, generate waveforms, calculate the amplitude and phase differences between them

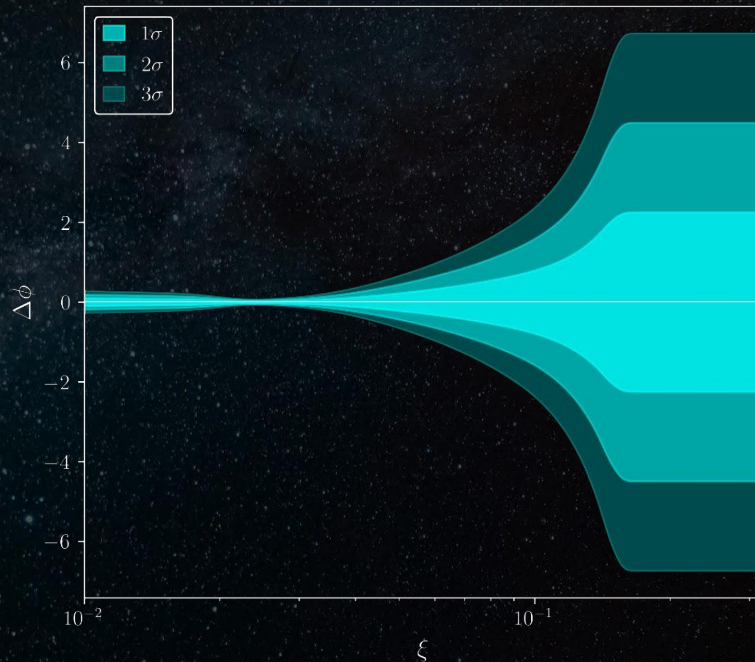


Correction Parameter Priors

- Range of differences between models



$$\sigma_{\mathcal{A}}(\xi) = \sqrt{(\overline{\Delta\mathcal{A}}(\xi))^2 + (\delta\mathcal{A}(\xi))^2}$$



$$\sigma_{\phi}(\xi) = \sqrt{(\overline{\Delta\phi}(\xi))^2 + (\delta\phi(\xi))^2}$$

Normalized Parameter Priors

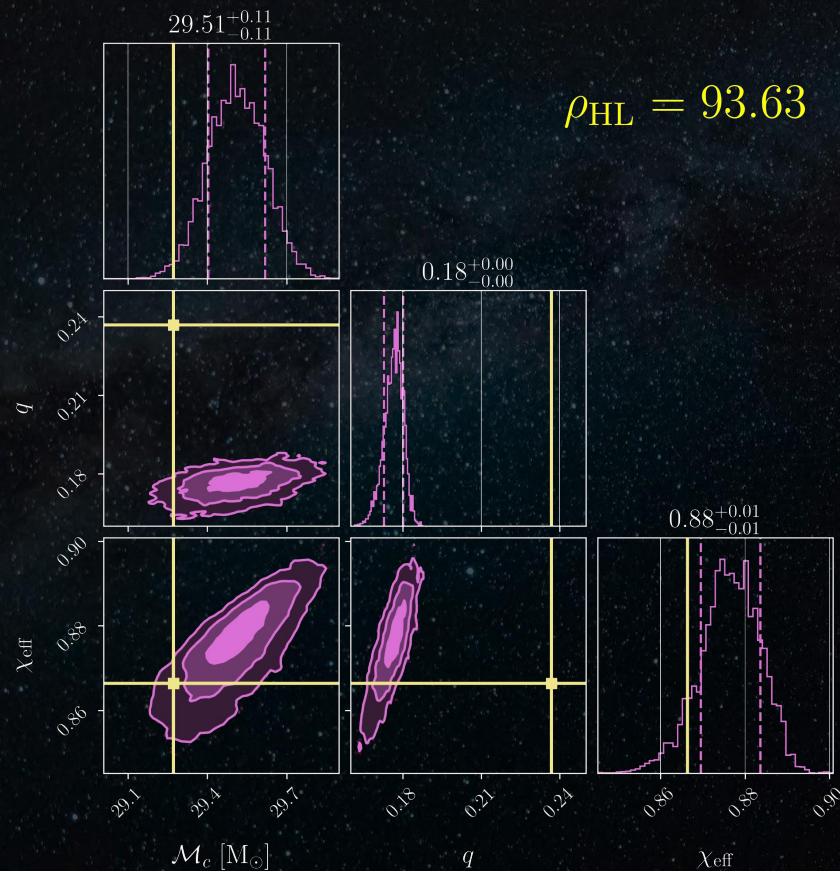
- We want to define Gaussian priors for our correction parameters:

$$\Pi(\alpha_k, \varphi_k) = \mathcal{N}(0, \sigma_{\mathcal{A}, \phi}(\xi_k))$$

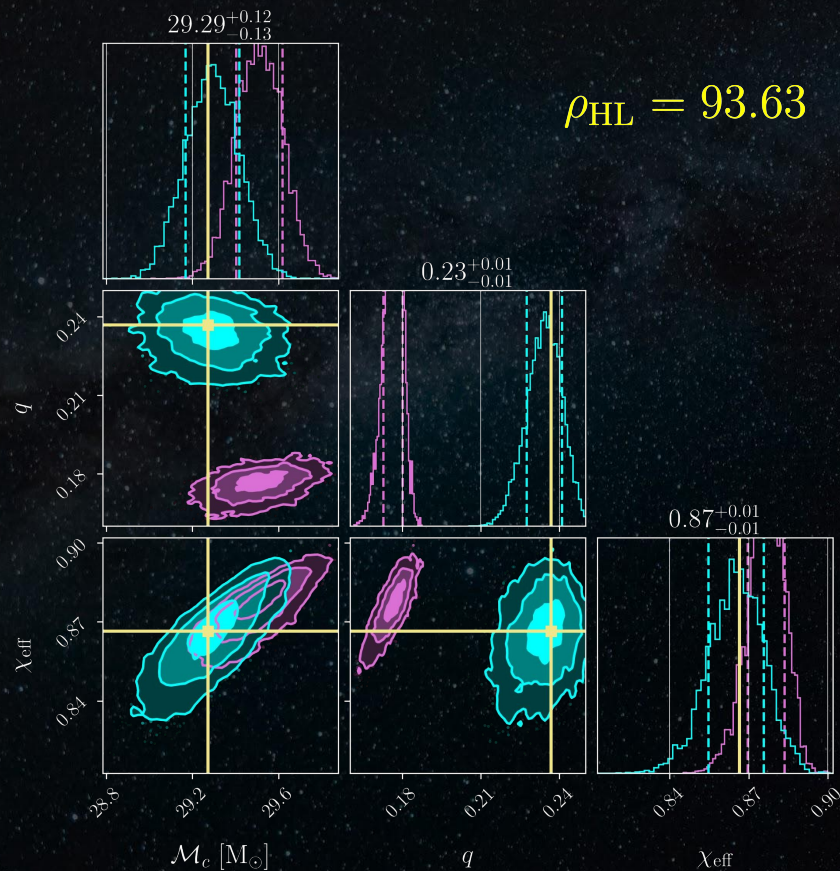
- Multiply the draw of a (0,1) Gaussian distribution by a number to change the standard deviation to that number!

$$\alpha_k = \sigma_{\mathcal{A}}(\xi_k) \tilde{\alpha}_k \quad \varphi_k = \sigma_{\phi}(\xi_k) \tilde{\varphi}_k$$

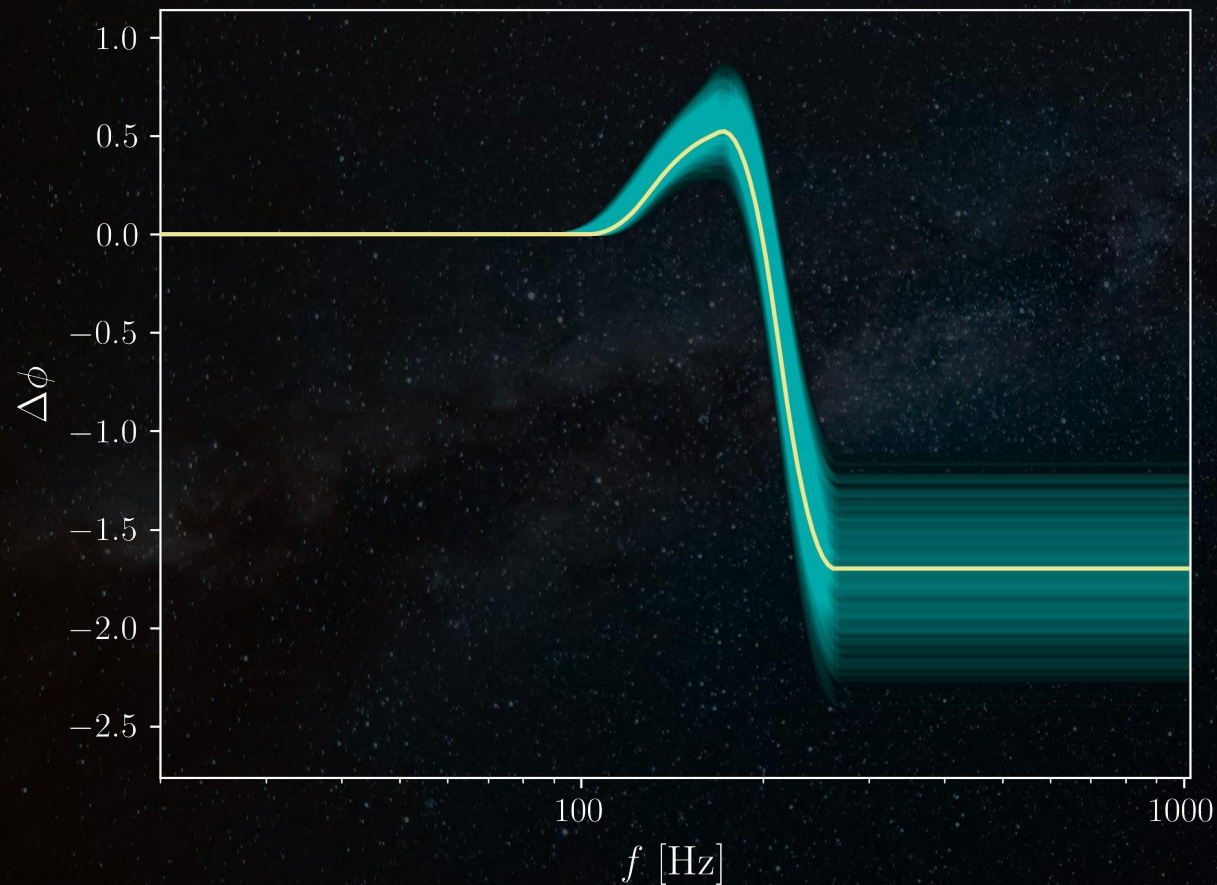
Unbiased Parameter Recovery



Unbiased Parameter Recovery



Recovered Waveform Correction



$$\Phi = \begin{cases} \tilde{\varphi}_0 = 0 \\ \tilde{\varphi}_1 = 1.17 \\ \tilde{\varphi}_2 = 1.05 \\ \tilde{\varphi}_3 = -1.98 \\ \xi_0 = 0.05 \\ \delta\tilde{\xi} = 0.20 \\ M = 89.65 M_{\odot} \end{cases}$$

Implications

- Run parameters plausible to be detected by LIGO
 - SNR comparable to [GW250114](#)
 - 93.63 vs 80 [Abac et. al. Phys. Rev. Lett. 135, 111403]
 - Applicable for detecting and correcting waveform systematics in real signals

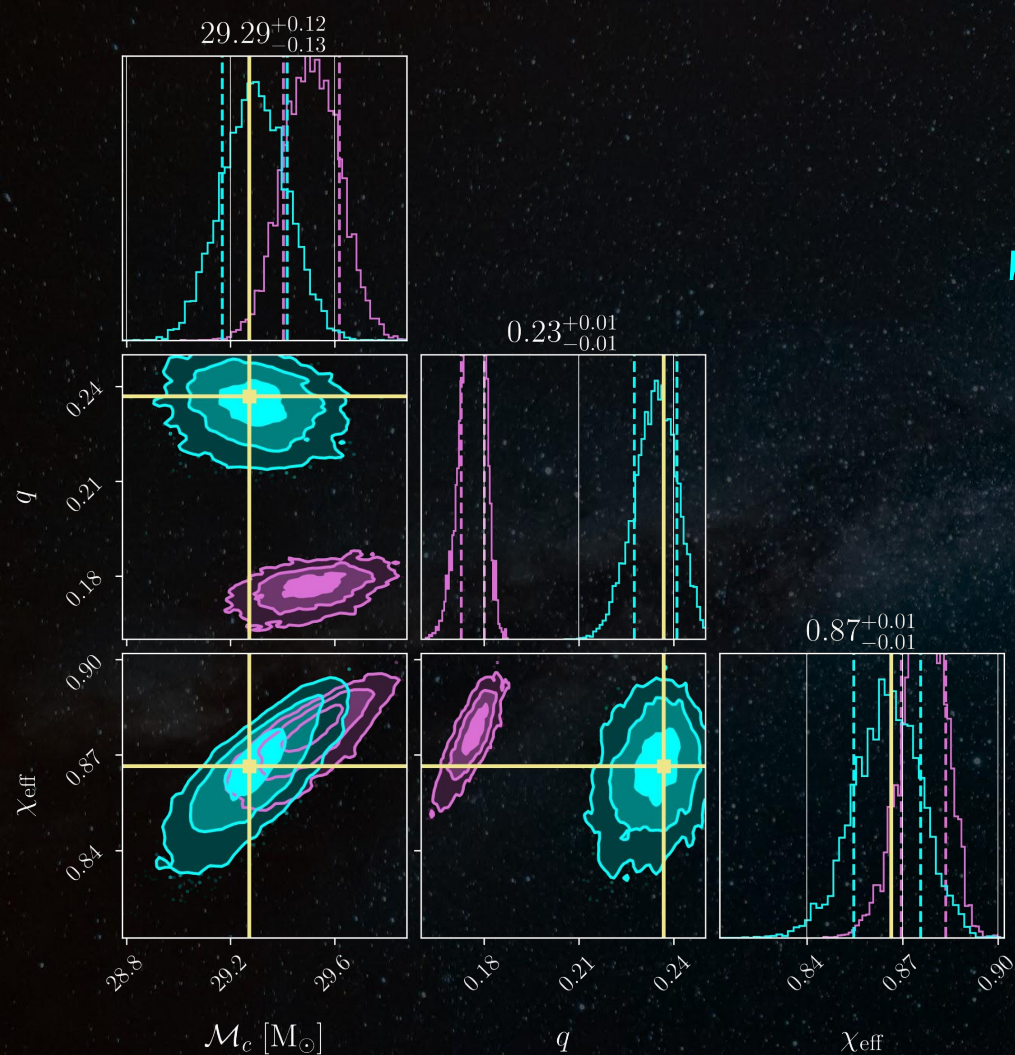
Problems

- Waveform model very slow
 - 1 hour $\rightarrow$ 30 hours

Future Work

- Examine real data with this model
 - Xu et. al. arXiv:2512.19513
- Combine with ppE Waveform Model
 - Bonilla et. al. Phys. Rev. D 107, 024015
 - Inspiral regime waveform correction

Thank you!



Acknowledgements

- Black Family Fellowship
- NSF PHYS-2409736
- Dr. Marceline Bonilla

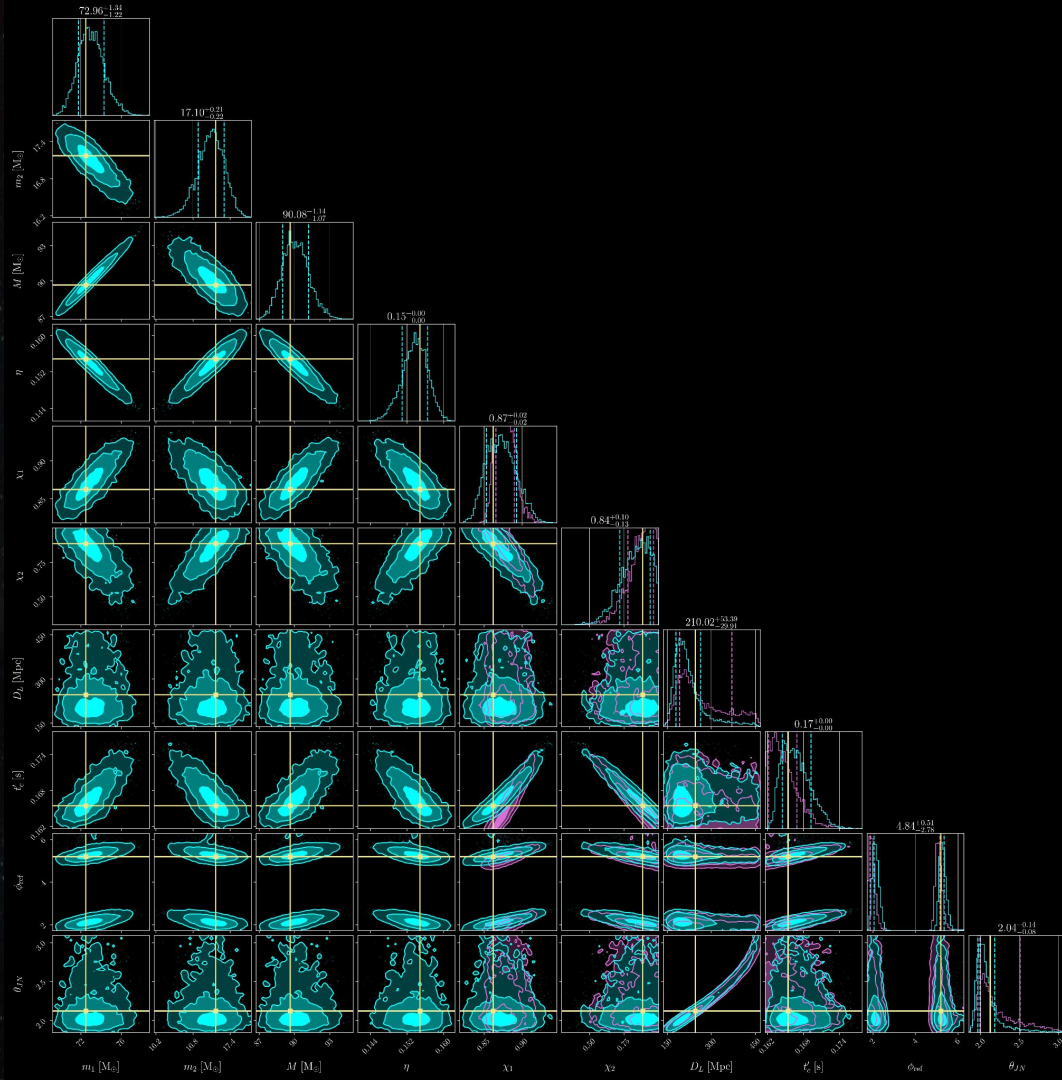
Frequency Node Constraint

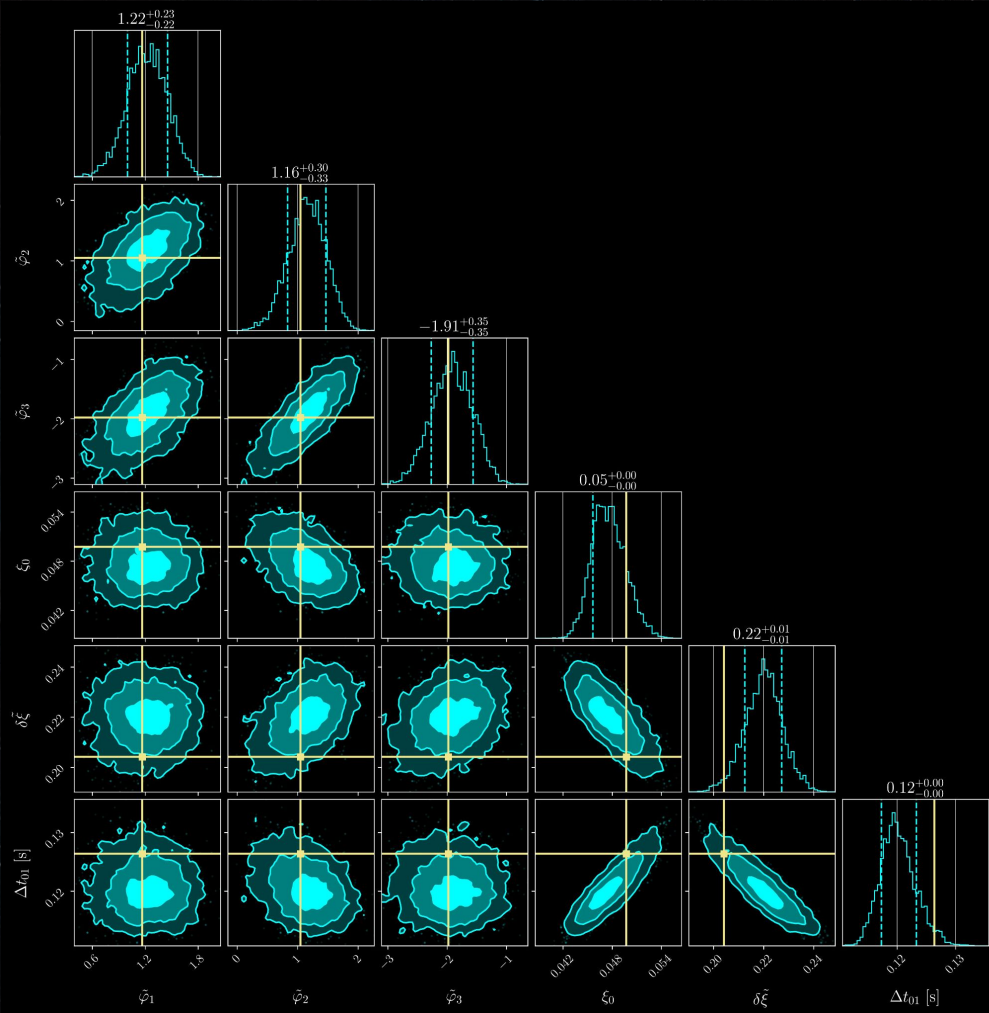
- The frequency band is discrete, so there is a limit to how close the frequency nodes can get to each other

$$\Delta f = 1/t_d$$

- Restrict the frequency nodes to be at least 4 frequency points apart
 - Edelman et. al. Phys. Rev. D, 103:042004

$$\Delta t_{01} \equiv \frac{1}{f_1 - f_0} \leq \frac{t_d}{4}$$





$$\Delta t_{01} = \frac{GM}{c^3 \xi_0} \left[\left(1 + \left[\frac{\xi_{\max} - \xi_0}{\xi_0} \right] \delta \tilde{\xi} \right)^{1/n} - 1 \right]^{-1}$$

$$\frac{c^3 \xi_{\max}}{G f_{\max}} \leq M \leq \frac{c^3 \xi_{\min}}{G f_{\min}} \qquad \varepsilon_{\alpha} \equiv 1 - \frac{1}{n} \sum_{k=1}^n \theta(\alpha_k + 1) \leq 1/2n$$